University of Engineering and Technology

2026-S-CYS-12

Anaya Azam

CYS – Section A

CMPE - 211L Database Systems

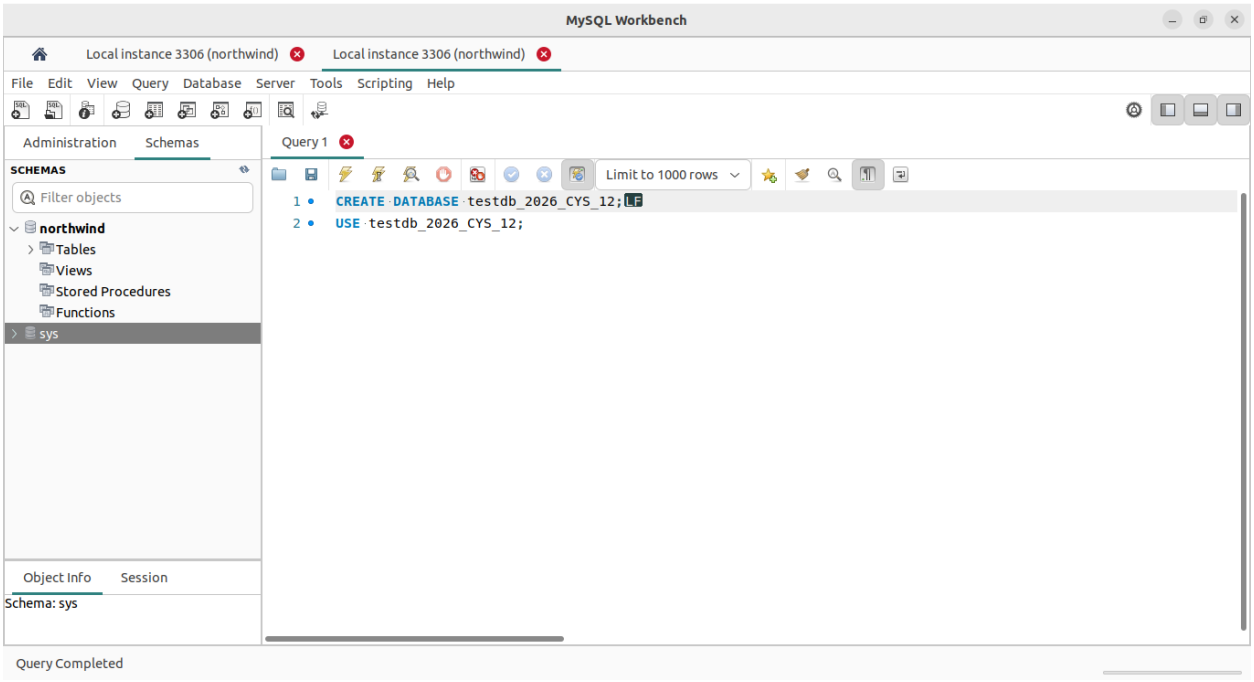
Lab – 01

Due: 23d September, 2026.

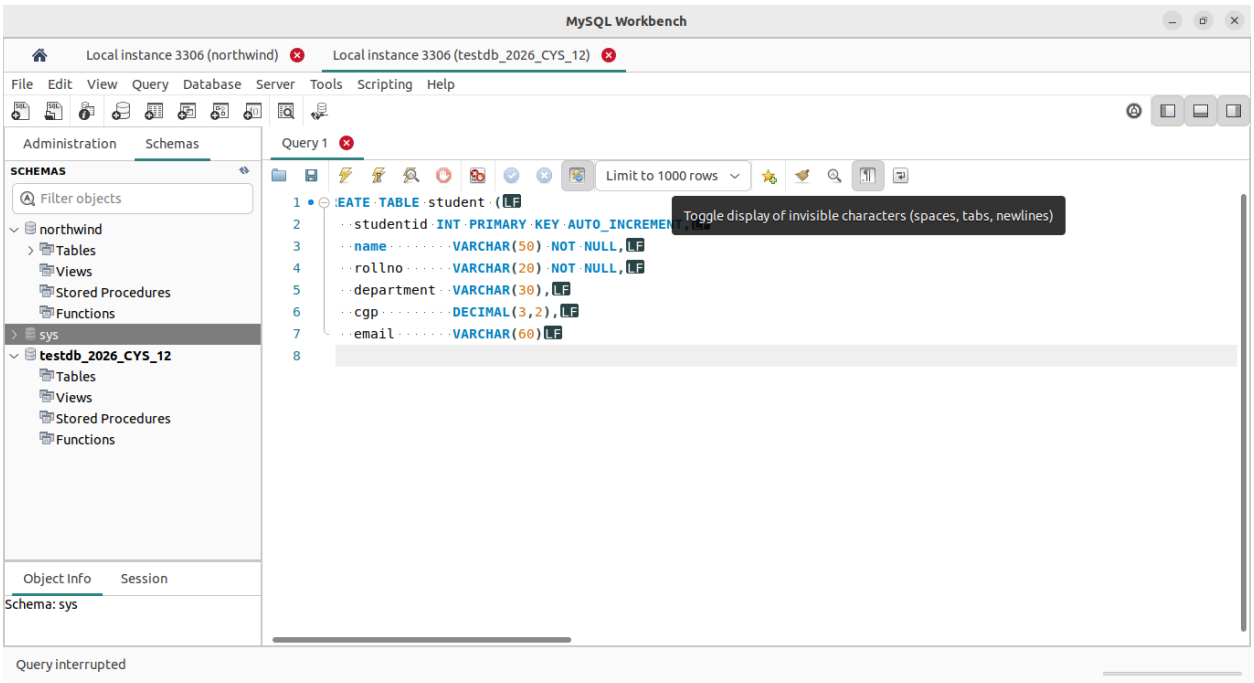
Lab Tasks

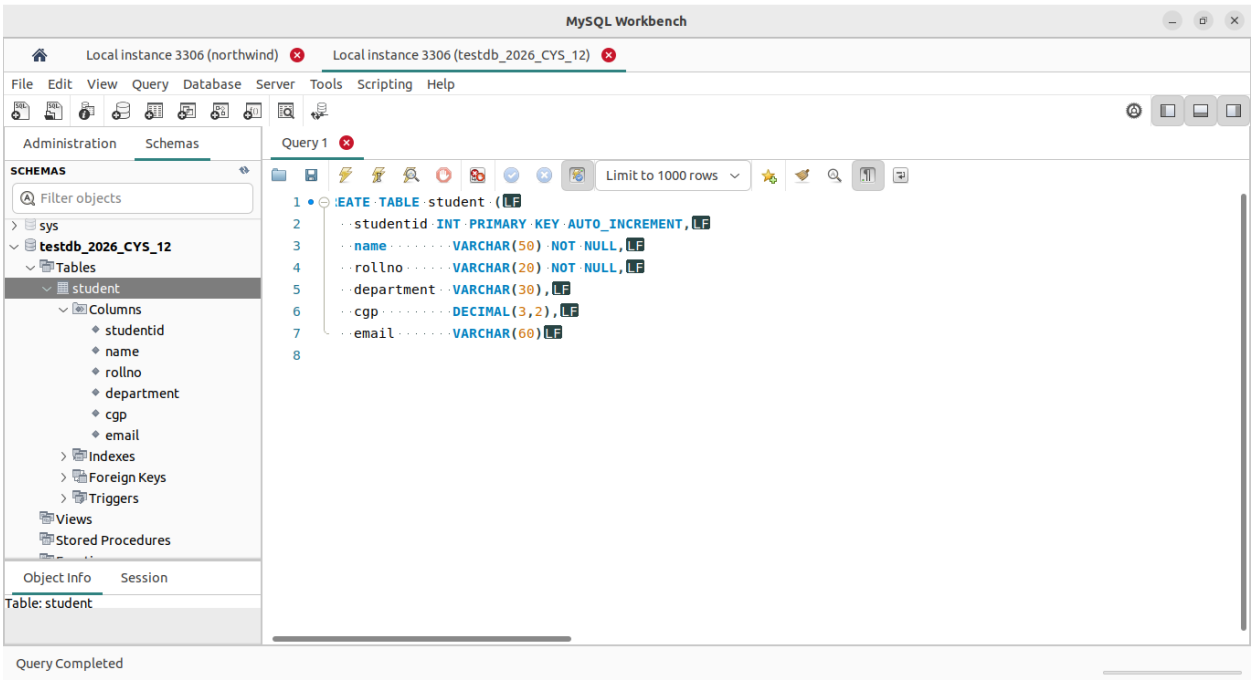
Part 1:

1. Create a new database named TestDB_2024_CS_X

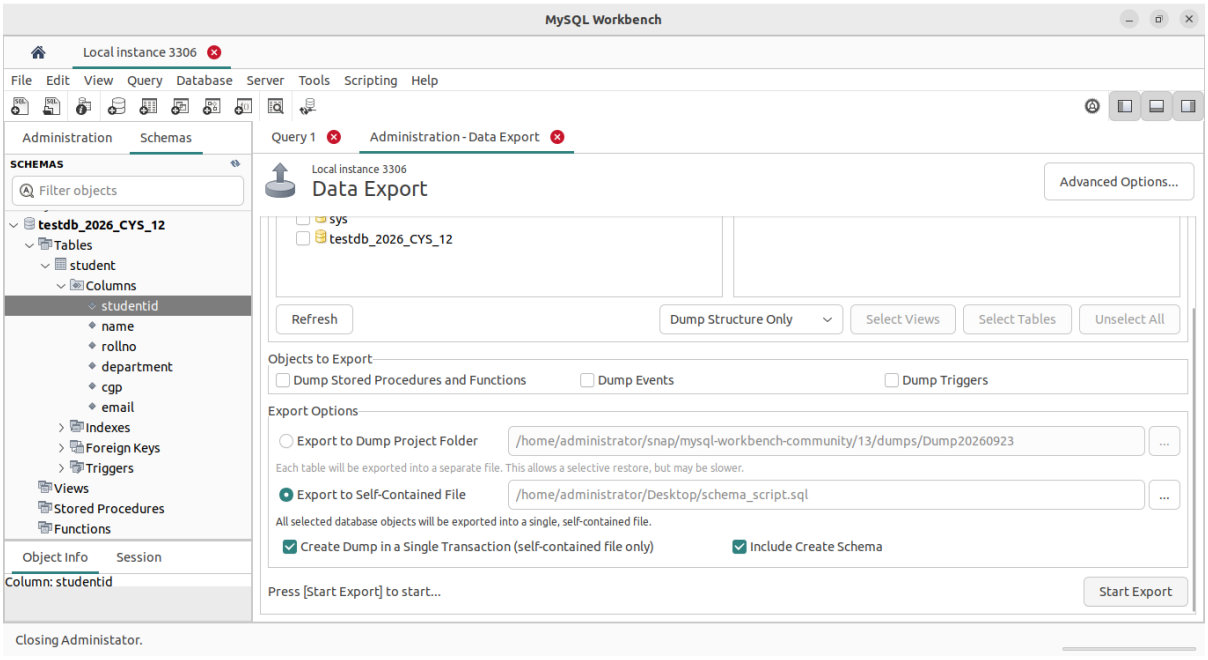


2. Create a new table named student using 5 to 6 attributes





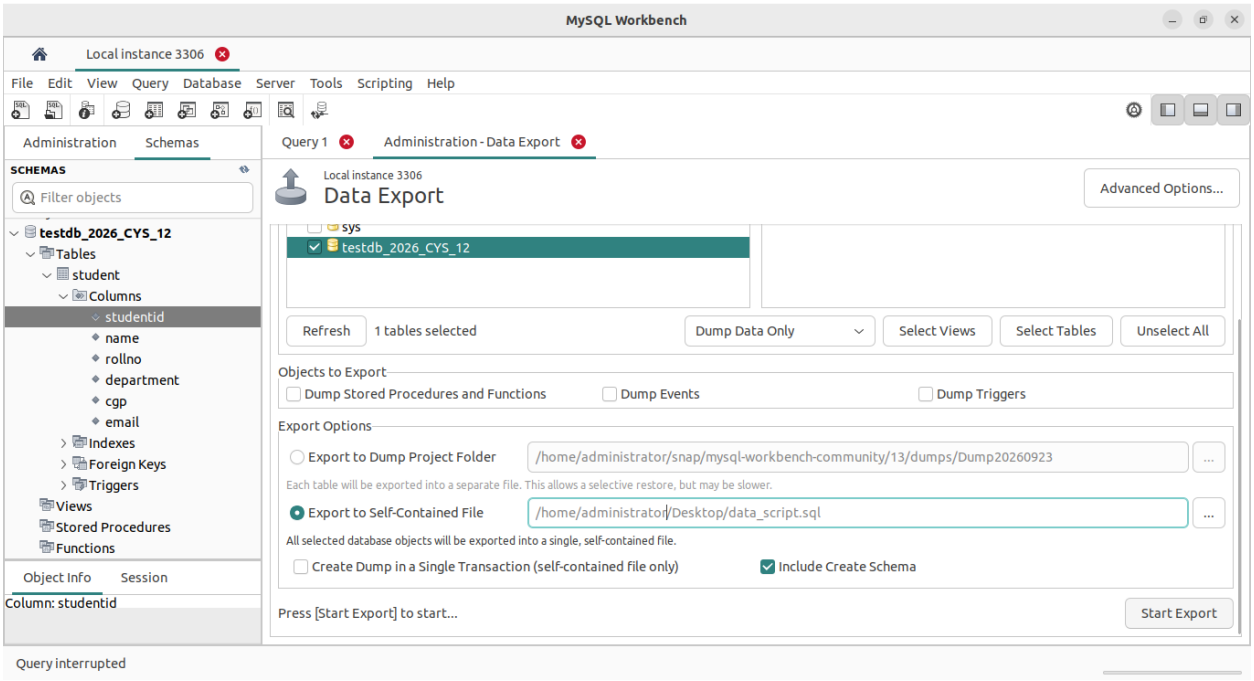
3. *Generate Scripts of your database*



4. *Restore your schema to another machine*



5. *Generate scripts of data*



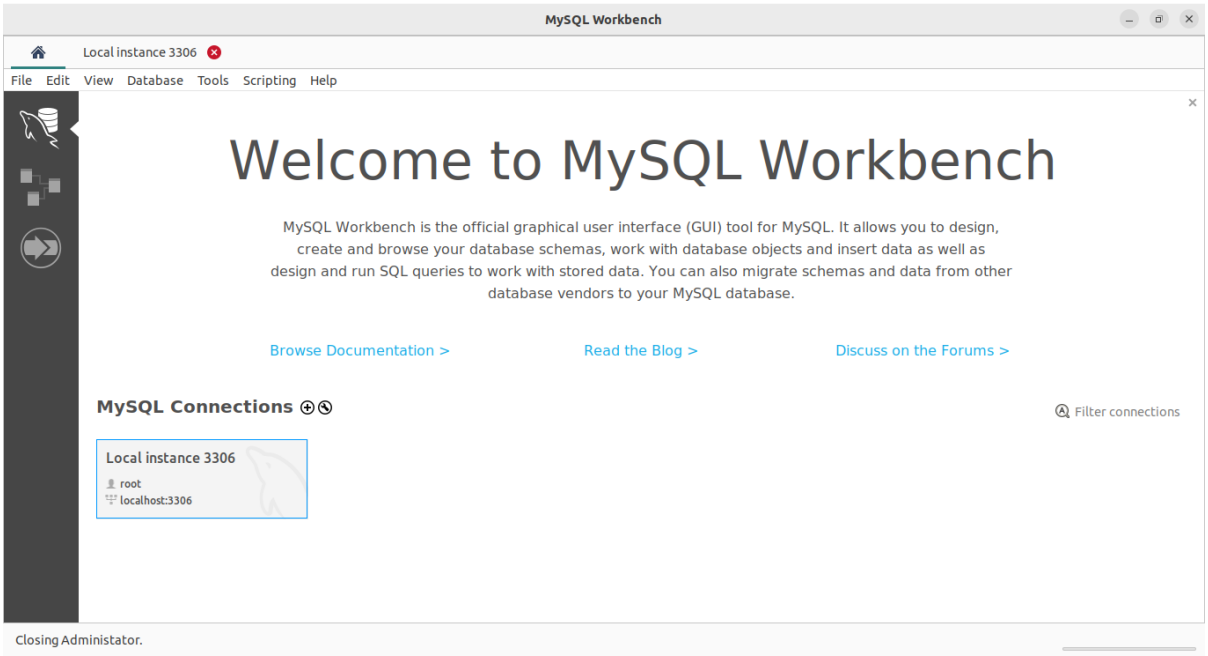
6. *Restore data to other system as well*

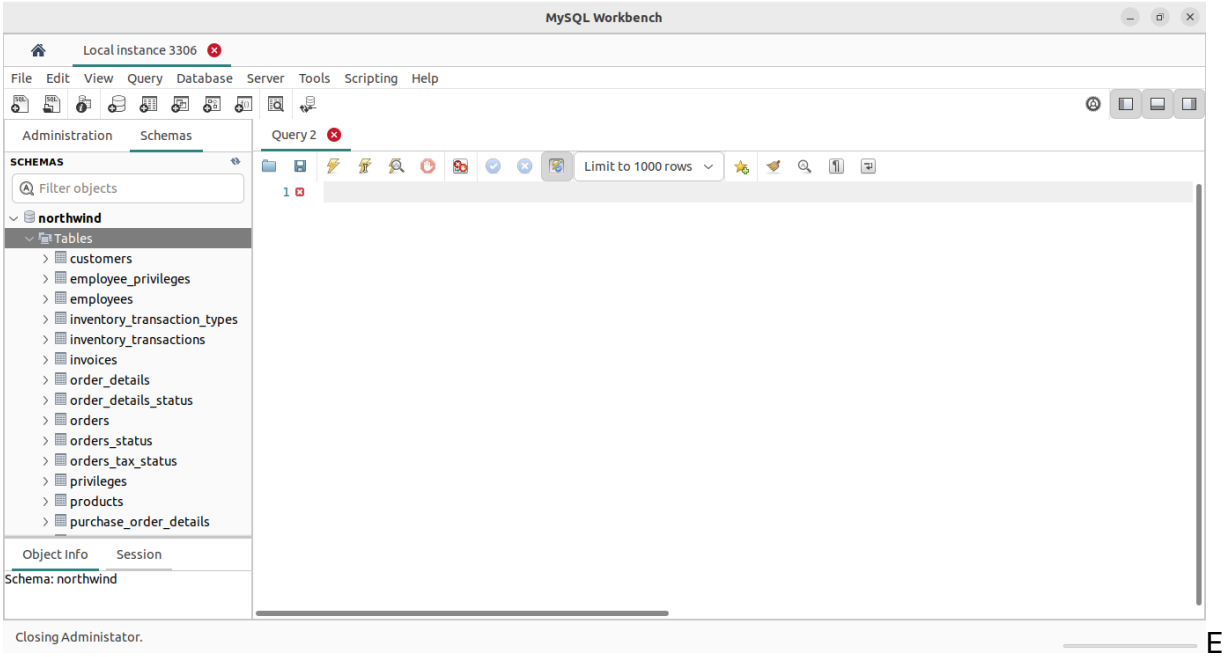


Part 2:

Task 1: Connect to the Northwind Database

• *Objective: Ensure the database connection is set up properly.*





E

Task 2: List All Tables

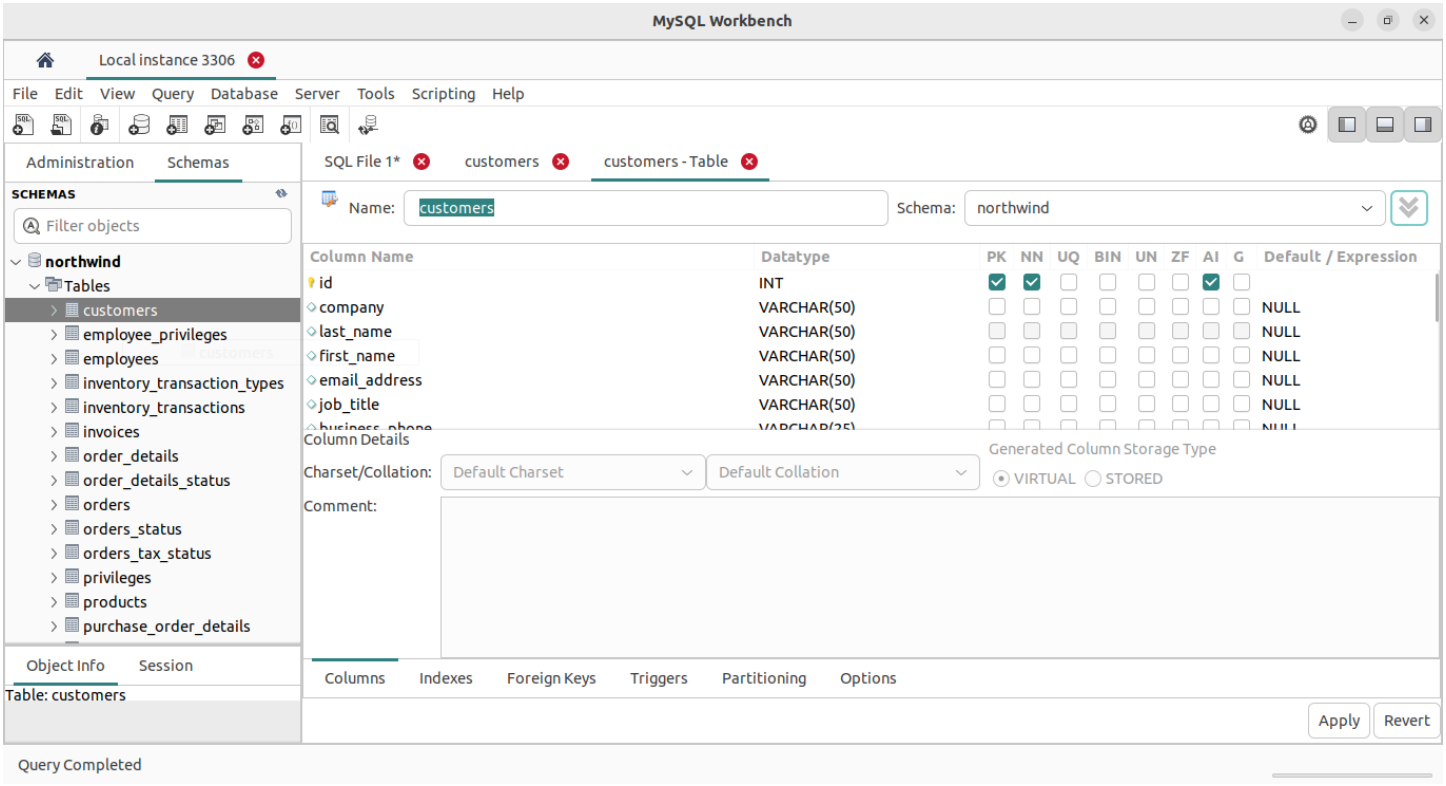
- *Objective: Identify the tables in the Northwind database.*

    Tables in northwind:

    - *customers*
    - *employee_privileges*
    - *employees*
    - *inventory_transaction_types*
    - *inventory_transactions*
    - *invoices*
    - *order_details*
    - *order_details_status*
    - *orders*
    - *orders_status*
    - *orders_tax_status*
    - *privileges*
    - *products*
    - *purchase_order_details*
    - *purchase_order_status*
    - *purchase_orders*
    - *sales_reports*
    - *shippers*
    - *strings*
    - *suppliers*

Task 3: View Table Structures

• *Objective: Understand the columns and data types in key tables.*



Column Name	Datatype	PK	NN	UQ	BIN	UN	ZF	AI	G	Default / Expression
id	INT	☑	☑	☐	☐	☐	☐	☑	☐	
company	VARCHAR(50)	☐	☐	☐	☐	☐	☐	☐	☐	NULL
last_name	VARCHAR(50)	☐	☐	☐	☐	☐	☐	☐	☐	NULL
first_name	VARCHAR(50)	☐	☐	☐	☐	☐	☐	☐	☐	NULL
email_address	VARCHAR(50)	☐	☐	☐	☐	☐	☐	☐	☐	NULL
job_title	VARCHAR(50)	☐	☐	☐	☐	☐	☐	☐	☐	NULL

Column Name	Datatype	PK	NN	UQ	BIN	UN	ZF	AI	G	Default / Expression
state_province	VARCHAR(50)	☐	☐	☐	☐	☐	☐	☐	☐	NULL
zip_postal_code	VARCHAR(15)	☐	☐	☐	☐	☐	☐	☐	☐	NULL
country_region	VARCHAR(50)	☐	☐	☐	☐	☐	☐	☐	☐	NULL
web_page	LONGTEXT	☐	☐	☐	☐	☐	☐	☐	☐	NULL
notes	LONGTEXT	☐	☐	☐	☐	☐	☐	☐	☐	NULL
attachments	LONGBLOB	☐	☐	☐	☐	☐	☐	☐	☐	NULL

Column Name	Datatype	PK	NN	UQ	BIN	UN	ZF	AI	G	Default / Expression
business_phone	VARCHAR(25)	☐	☐	☐	☐	☐	☐	☐	☐	NULL
home_phone	VARCHAR(25)	☐	☐	☐	☐	☐	☐	☐	☐	NULL
mobile_phone	VARCHAR(25)	☐	☐	☐	☐	☐	☐	☐	☐	NULL
fax_number	VARCHAR(25)	☐	☐	☐	☐	☐	☐	☐	☐	NULL
address	LONGTEXT	☐	☐	☐	☐	☐	☐	☐	☐	NULL
city	VARCHAR(50)	☐	☐	☐	☐	☐	☐	☐	☐	NULL

Task 4: Record Key Details

• *Objective: Document the structure of a few key tables.*

FOR CUSTOMERS:

Column Name	*Data Type*	*Key/Constraint*
id	*INT*	*PK, NN, AI*
company	*VARCHAR(50)*	--
last_name	*VARCHAR(50)*	--
first_name	*VARCHAR(50)*	--
email_ address	*VARCHAR(50)*	--
jobtitle	*VARCHAR(50)*	--

FOR SALES_REPORT:

Column Name	*Data Type*	*Key/Constraint*
group_by	*VARCHAR(50)*	*PK, NN*
display	*VARCHAR(50)*	--
title	*VARCHAR(50)*	--
*filter_row*_source	LONGTEXT	--
def*ault*	*TINYINT(1)*	*NOT NULL*

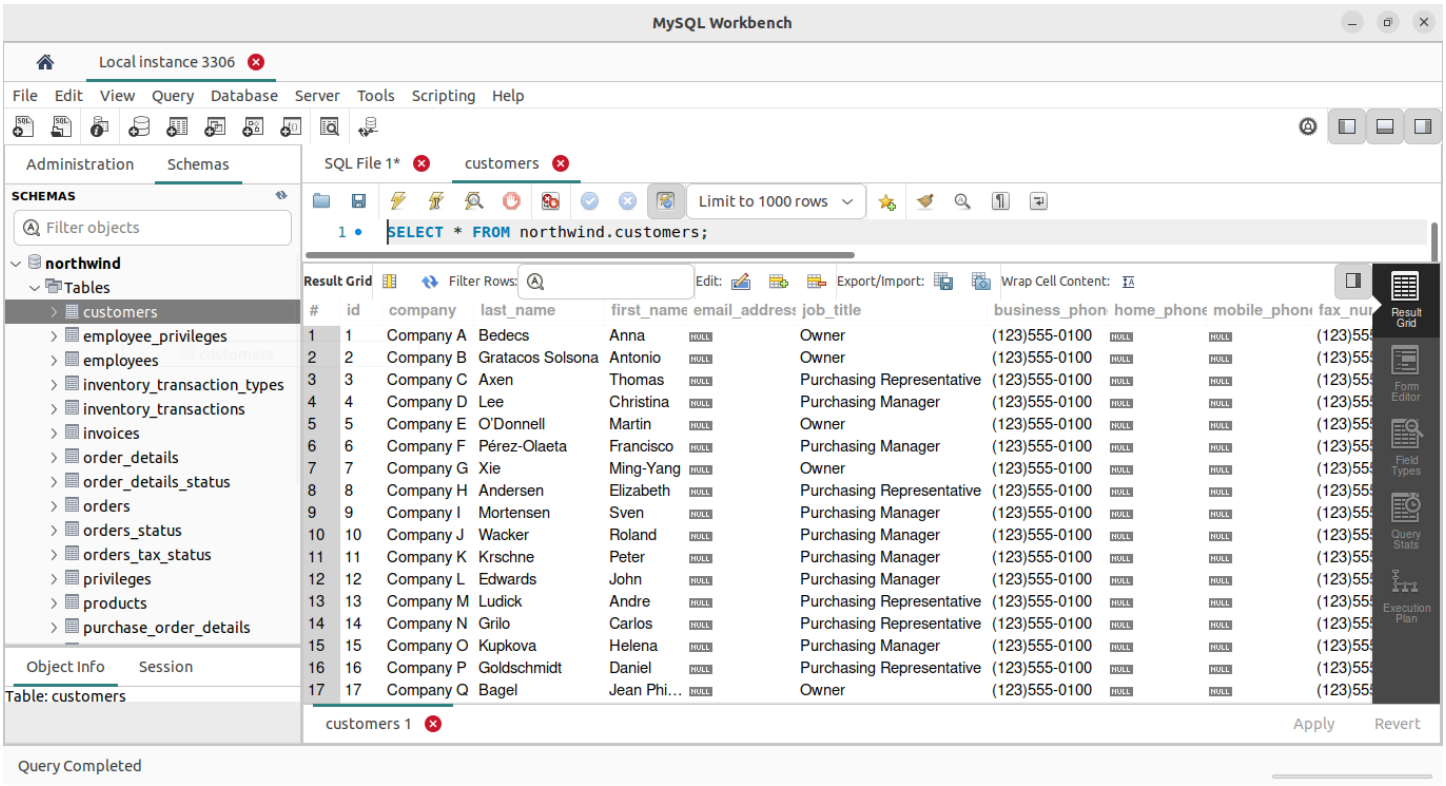
FROM PRODUCTS:

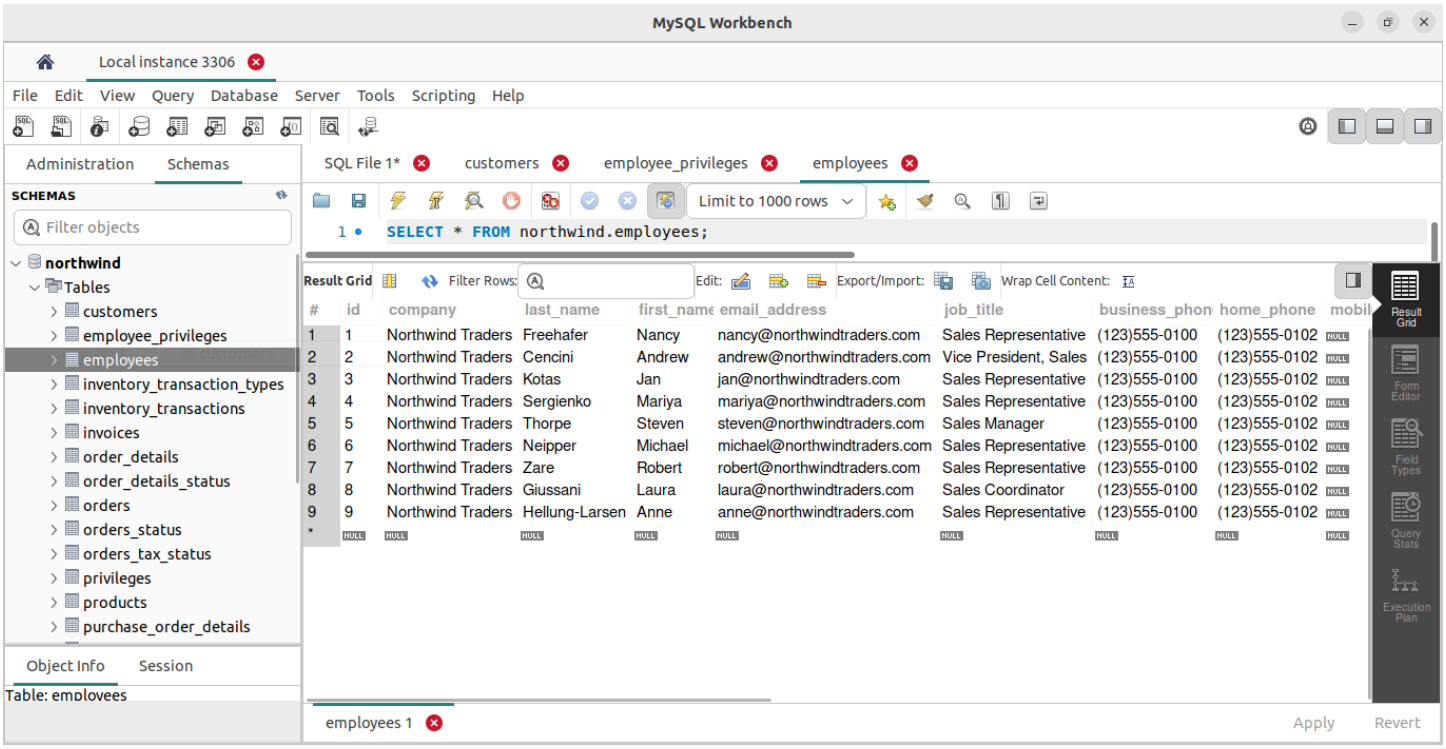
Column Name	*Data Type*	*Key/Constraint*
supplier_ids	*LONGTEXT*	--
id	*INT*	*PK,NN, I*
product_code	*VARCHAR(25)*	--
product_name	*VARCHAR(50)*	*NOT NULL*
list_price	*DECIMAL(19,4)*	*NOT NULL*

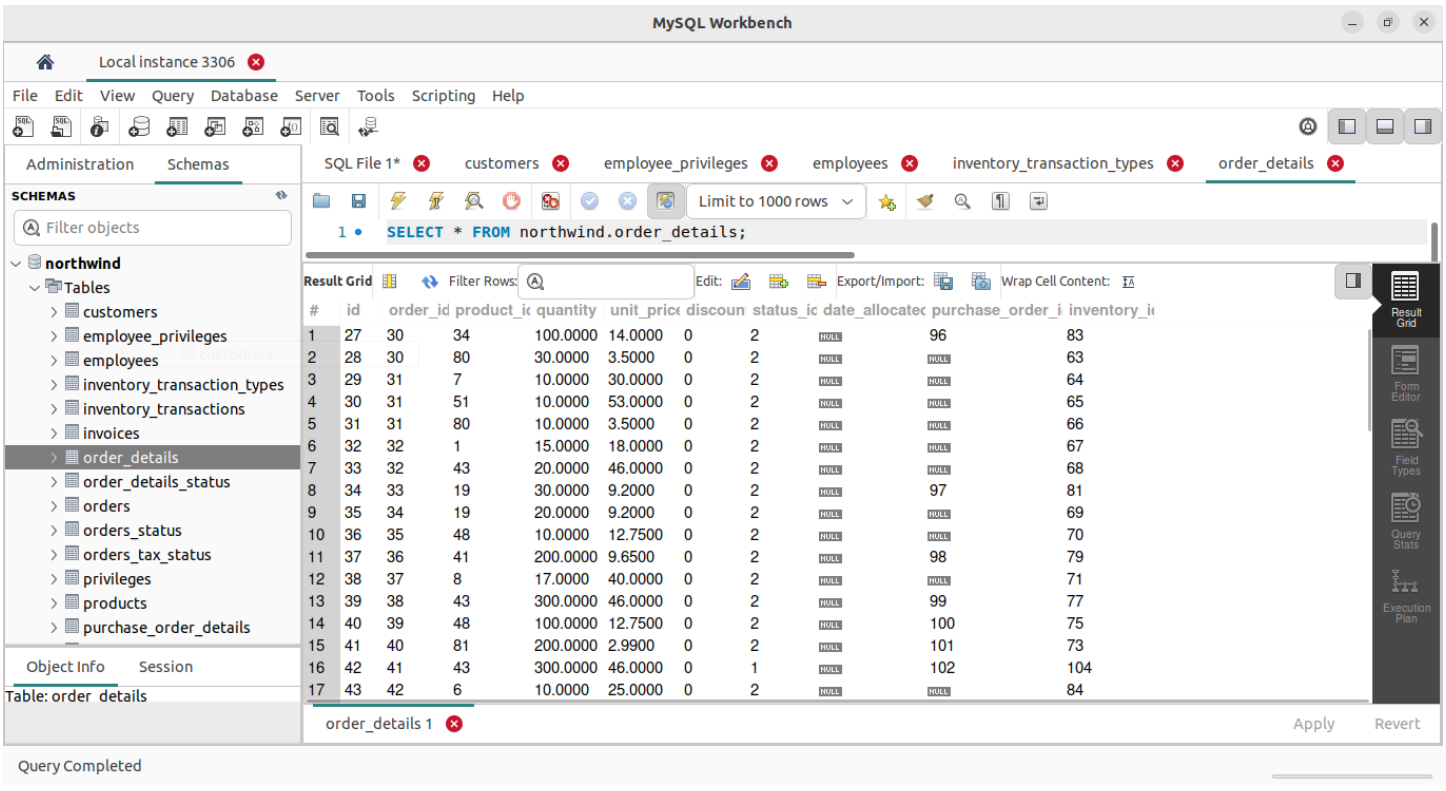
discontinued	*TINYINT(1)*	*NOT NULL*

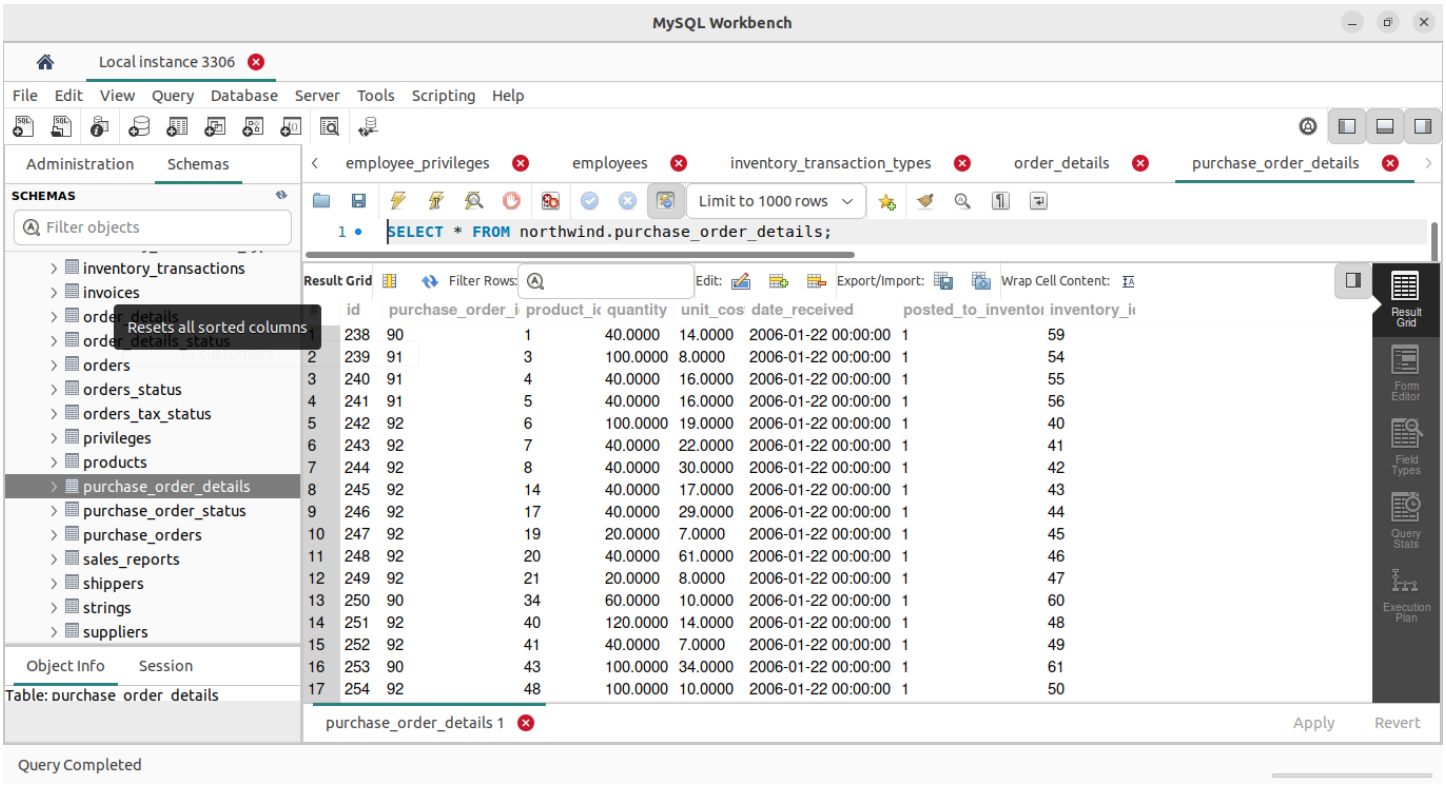
Task 5: Browse Data

- *Objective: View a sample of the data stored in each table.*




MySQL Workbench
Local instance 3306
File Edit View Query Database Server Tools Scripting Help
Administration Schemas
SCHEMAS
Filter objects
northwind
Tables
customers
employee_privileges
employees
inventory_transaction_types
inventory_transactions
invoices
order_details
order_details_status
orders
orders_status
orders_tax_status
privileges
products
purchase_order_details
Object Info Session
Table: employees
SQL File 1* customers employee_privileges employees
Limit to 1000 rows
1 • SELECT * FROM northwind.employees;
Result Grid Filter Rows: Edit: Export/Import: Wrap Cell Content:
id company last_name first_name email_address job_title business_phon home_phone mobil
1 1 Northwind Traders Freehafer Nancy nancy@northwindtraders.com Sales Representative (123)555-0100 (123)555-0102 NULL
2 2 Northwind Traders Cencini Andrew andrew@northwindtraders.com Vice President, Sales (123)555-0100 (123)555-0102 NULL
3 3 Northwind Traders Kotas Jan jan@northwindtraders.com Sales Representative (123)555-0100 (123)555-0102 NULL
4 4 Northwind Traders Sergienko Mariya mariya@northwindtraders.com Sales Representative (123)555-0100 (123)555-0102 NULL
5 5 Northwind Traders Thorpe Steven steven@northwindtraders.com Sales Manager (123)555-0100 (123)555-0102 NULL
6 6 Northwind Traders Neipper Michael michael@northwindtraders.com Sales Representative (123)555-0100 (123)555-0102 NULL
7 7 Northwind Traders Zare Robert robert@northwindtraders.com Sales Representative (123)555-0100 (123)555-0102 NULL
8 8 Northwind Traders Giussani Laura laura@northwindtraders.com Sales Coordinator (123)555-0100 (123)555-0102 NULL
9 9 Northwind Traders Hellung-Larsen Anne anne@northwindtraders.com Sales Representative (123)555-0100 (123)555-0102 NULL
Result Grid
Form Editor
Field Types
Query Stats
Execution Plan
employees 1
Apply Revert
Query Completed


MySQL Workbench
Local instance 3306
File Edit View Query Database Server Tools Scripting Help
Administration Schemas
SCHEMAS
Filter objects
northwind
Tables
customers
employee_privileges
employees
inventory_transaction_types
inventory_transactions
invoices
order_details
order_details_status
orders
orders_status
orders_tax_status
privileges
products
purchase_order_details
Object Info Session
Table: order_details
SQL File 1* customers employee_privileges employees inventory_transaction_types order_details
Limit to 1000 rows
1 • SELECT * FROM northwind.order_details;
Result Grid Filter Rows: Edit: Export/Import: Wrap Cell Content:
id order_id product_id quantity unit_price discoun status_id date_allocated purchase_order_i inventory_i
1 27 30 34 100.0000 14.0000 0 2 NULL 96 83
2 28 30 80 30.0000 3.5000 0 2 NULL NULL 63
3 29 31 7 10.0000 30.0000 0 2 NULL NULL 64
4 30 31 51 10.0000 53.0000 0 2 NULL NULL 65
5 31 31 80 10.0000 3.5000 0 2 NULL NULL 66
6 32 32 1 15.0000 18.0000 0 2 NULL NULL 67
7 33 32 43 20.0000 46.0000 0 2 NULL NULL 68
8 34 33 19 30.0000 9.2000 0 2 NULL 97 81
9 35 34 19 20.0000 9.2000 0 2 NULL NULL 69
10 36 35 48 10.0000 12.7500 0 2 NULL NULL 70
11 37 36 41 200.0000 9.6500 0 2 NULL 98 79
12 38 37 8 17.0000 40.0000 0 2 NULL NULL 71
13 39 38 43 300.0000 46.0000 0 2 NULL 99 77
14 40 39 48 100.0000 12.7500 0 2 NULL 100 75
15 41 40 81 200.0000 2.9900 0 2 NULL 101 73
16 42 41 43 300.0000 46.0000 0 1 NULL 102 104
17 43 42 6 10.0000 25.0000 0 2 NULL NULL 84
Result Grid
Form Editor
Field Types
Query Stats
Execution Plan
order_details 1
Apply Revert
Query Completed



Task 6: Document Examples

• Objective: Record a few rows of data from key tables for reference.

Employees:

id	*name*	*Job_title*	*city/state*	*country*
1	*Company A*	*Owner*	*Seattle,WA*	*USA*
2	*Company B*	*Owner*	*Boston.MA*	*USA*
3	*Company C*	*Purchasing representative*	*Los Angeles,CA*	*USA*

Products:

id	*Product_code*	*product_name*	*category*	*Standard_cost*	*List_price*
1	*NWTB-1*	*Nortwind traders Chai*	*Beverages*	*13.50*	*18.00*
2	*NWTCO-3*	*Northwing Trader syrup*	*Condiments*	*7.50*	*10.00*
3	*NWTCO-4*	*Nortwind traders Cajun seasoning*	*Condiments*	*16.50*	*22.00*

Task 7: Review Table Relationships

- *Customers → Orders → Order_Details → Products — a customer places orders, each order has line items, each line item is one product.*
- *Suppliers → Purchase_Orders → Purchase_Order_Details → Products — same pattern, but for restocking from suppliers.*
- *Employees are linked to both Orders and Purchase_Orders (they process sales and purchasing).*
- *Orders → Invoices — each order gets billed on an invoice.*

Task 8: Prepare a Lab Report

• Objective: Summarize findings in a concise report.

Tables in northwind: (20)

- *customers*
- *employee_privileges*
- *employees*
- *inventory_transaction_types*
- *inventory_transactions*
- *invoices*
- *order_details*
- *order_details_status*
- *orders*
- *orders_status*
- *orders_tax_status*
- *privileges*
- *products*
- *purchase_order_details*
- *purchase_order_status*
- *purchase_orders*
- *sales_reports*
- *shippers*
- *strings*
- *suppliers*